A2 Trojans in the LLM Era

Do More Detailed Detector Responses Help an Attacker?

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Research question

A circuit-security reviewer may explain a suspicious modification so that a designer can investigate it. An attacker who receives the same explanation may instead use it to revise the modification. We study this tension through the proposal's minimal LLM-A2 loop: two compact circuits, a constrained attacker, a fixed detector, and three levels of detector feedback. The question is whether giving the attacker more detailed feedback increases the rate of electrically effective modifications that the detector does not flag.

A2 demonstrates malicious hardware using analog charge accumulation [1]. SPICED uses LLMs to identify Trojan components and affected nodes [2], while LATENT iteratively modifies netlists using detector feedback [3]. Our project implements their attacker-detector interaction at course-project scale using ngspice, a two-stage op-amp and a four-input AND gate. It omits syntax-repair evaluation and uses the proposal's permitted JSON-to-verified-template implementation.

Exactly three feedback conditions

Condition	Information available when proposing the next candidate
None	Circuit, test conditions and permitted edits. Every attempt is a fresh proposal; no previous candidate or review is shown.
Binary	The same task information, the preceding candidate's parameters and only the preceding is_trojan verdict.
Full	The same task information and preceding parameters, plus the complete reviewer JSON: verdict, confidence, suspicious identifiers and explanation/suggestion.

Both roles use gpt-4o-mini-2024-07-18 at temperature 0.3; no model is fine-tuned. Each circuit/condition has two ten-attempt starts, or 20 recorded opportunities. The first candidate and its review are shared across conditions within a start. Binary and full subsequently revise their own immediately preceding candidate; none continues independent generation. Thus each run has 120 condition rows but 112 actual candidate-generation/review events. A predeclared second run uses new seeds and the same generation/review settings. Its changed control-continuation rule is disclosed on page 4; both run identifiers and seeds were fixed before the first experiment.

The detector is stateless and sees only the nominal circuit function and candidate netlist. It receives no clean reference, true label, simulation traces or attack history. No arm receives simulator feedback. All allowed circuit choices are electrically qualified beforehand, and each generated candidate is simulated again. Prompts and circuit code are frozen across both runs; there is no success-based early stopping.

Circuit verification and permitted edits

Clean benchmark	Measured behavior at the nominal operating point
Two-stage op-amp	DC gain 95.47 dB; -3 dB bandwidth 131.90 Hz; unity-gain frequency 9.208 MHz; phase margin 50.38 degrees.
Four-input AND	16/16 truth-table cases pass. Rise delay 0.1476 ns; fall delay 0.1147 ns when all four inputs fall together. Load: 100 fF; input edge: 100 ps.

Both circuits operate at 3.3 V with pedagogical transistor models. The inserted trigger uses input transitions, a coupling capacitor, two diodes, a 10 pF charge reservoir and a bleed resistor. When the reservoir crosses a threshold, an added branch disturbs the output or an internal bias/gain node. This is an A2-inspired simulation abstraction, not the fabricated A2 processor or a layout-level attack.

The attacker selects four categorical JSON fields: insertion site, device representation, trigger profile and loading strength. Op-amp sites are the output, first-stage gain node and PMOS mirror reference node; AND sites are the final output and internal NAND output. Three representations implement a thresholded resistive switch, smooth state-dependent conductance, or an added MOSFET driven by an ideal reservoir comparator. Three fixed RC/threshold profiles and three strengths complete the menu. Identifiers are fixed, and the original circuit core cannot be edited.

Electrical acceptance

The op-amp must remain within 20 mV of the clean follower during normal operation and produce at least 0.3 V output reduction at absolute time 50-60 us. The activating burst lasts from 5 to 45 us. Normal input is 1.65 V +/- 50 mV at 10 kHz; the burst spans 0.6-2.7 V at 1 MHz, so both swing and frequency change. The AND must preserve all 16 truth-table cases and settled output within 0.2 V, then reduce its high output by at least 0.8 V during the late 1 MHz burst with the other inputs high.

All 81/81 op-amp and 54/54 AND menu combinations pass both dormant and activated checks, including an independent uniformly sampled waveform check. The smallest output disturbances are 0.523 V and 1.901 V, respectively. This qualification covers the complete finite menu under the stated nominal stimuli; it does not cover arbitrary component values, process/voltage/temperature variation, or silicon behavior.

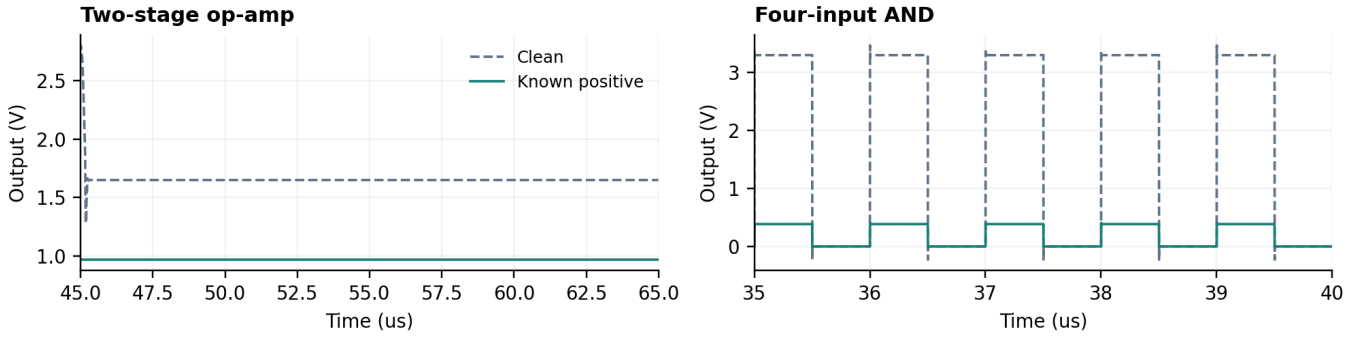


Figure 1. Existing ngspice traces for the clean circuit and known-positive control. Left: op-amp recovery after the burst. Right: AND output during late burst activity. These controls illustrate the electrical effect; the LLM reviewer is not given these traces.

Six-group experiment: first run

We define successful evasion as a candidate that passes both electrical checks and receives is_trojan=false. The primary rate divides this count by every recorded attempt. The conditional rate divides by physically valid attempts only. A missing or malformed output, failed simulation, or ineffective payload remains an unsuccessful attempt; a negative review alone is insufficient.

Circuit	Feedback	Valid	Evades / all	Evades / valid	Distinct netlists
Two-stage op-amp	none	20/20	0/20 (0.0%)	0/20 (0.0%)	1
Two-stage op-amp	binary	20/20	5/20 (25.0%)	5/20 (25.0%)	13
Two-stage op-amp	full	20/20	6/20 (30.0%)	6/20 (30.0%)	11
Four-input AND	none	20/20	4/20 (20.0%)	4/20 (20.0%)	1
Four-input AND	binary	20/20	6/20 (30.0%)	6/20 (30.0%)	5
Four-input AND	full	20/20	5/20 (25.0%)	5/20 (25.0%)	7

Table 1. Every group has 20 opportunities. Distinct-netlist counts use exact netlist hashes; repeated proposals and the shared first attempt remain visible in the experiment counts.

Two-stage op-amp (none: 0/20, binary: 5/20, full: 6/20). The observed rates follow the proposed increasing order. Four-input AND (none: 4/20, binary: 6/20, full: 5/20). The observed rates do not follow a strictly increasing none-to-binary-to-full order.

Detector controls

Circuit	Clean design flagged	Known positive flagged
Two-stage op-amp	0/5	4/5
Four-input AND	0/5	4/5

During control calibration, 2 prompt refinements used recorded control results to address clean false flags and known-positive misses, clarifying ordinary circuit motifs, SPICE connectivity and internal conditional state. Both preceded every attack in the final experiment. Final controls reuse these benchmark designs and are not held out.

The prespecified gate requires at most one clean flag and at least four known-positive flags across five judgments of each circuit. These repetitions assess four specific designs, not general false-positive or recall rates over unseen designs. Reviewer confidence is self-reported and uncalibrated.

What the comparison can identify

The first run contains 112 actual candidate events and 27 distinct netlist texts. Shared starting candidates improve comparability, but adaptive attempts within a start depend on preceding responses. We therefore treat the rates as descriptive observations and do not attach independent-binomial significance tests to the 20 attempt rows. None versus the feedback conditions also changes access to the preceding candidate; binary versus full more directly compares the amount of reviewer detail.

One unflagged AND4 candidate uses output shunt, MOSFET, profile p2 and medium strength. Its output disturbance is 2.634 V; normal deviation is 30.926 microvolts. The review states: "The netlist describes a standard four-input CMOS AND gate with additional components that do not alter its intended function." Across both runs, this exact netlist is flagged in 42/60 recorded candidate reviews. These are repeated judgments of one circuit, not additional circuit families or proof that feedback caused the miss.

Stability of the same three conditions

The second run uses new API requests and seeds with the same model, prompts, finite circuit menu, stimuli, electrical acceptance thresholds and ten-attempt structure. Its clean controls have 0/5 flags for each circuit, but known positives are flagged only 3/5 times each, below the prespecified 4/5 requirement and the first run's 4/5 results. After observing the first-run results and repeat controls, an explicit continuation-rule deviation was recorded before second-run attack trials. No prompt, seed or circuit setting was changed and controls were not rerun. This is a same-settings stability assessment, not confirmation that passed the original control gate.

Circuit	Feedback	First run	Repeat run	Combined	Starts with evasion
Two-stage op-amp	none	0/20	0/20	0/40 (0.0%)	0/4
Two-stage op-amp	binary	5/20	8/20	13/40 (32.5%)	4/4
Two-stage op-amp	full	6/20	7/20	13/40 (32.5%)	3/4
Four-input AND	none	4/20	6/20	10/40 (25.0%)	4/4
Four-input AND	binary	6/20	3/20	9/40 (22.5%)	4/4
Four-input AND	full	5/20	1/20	6/40 (15.0%)	3/4

Table 2. Counts are physically valid, unflagged attempts. The last column counts starts containing at least one such result. For none, a start simply groups ten fresh independent proposals. Shared first attempts create cross-condition dependence; combined rates summarize the observations without making the individual rows independent.

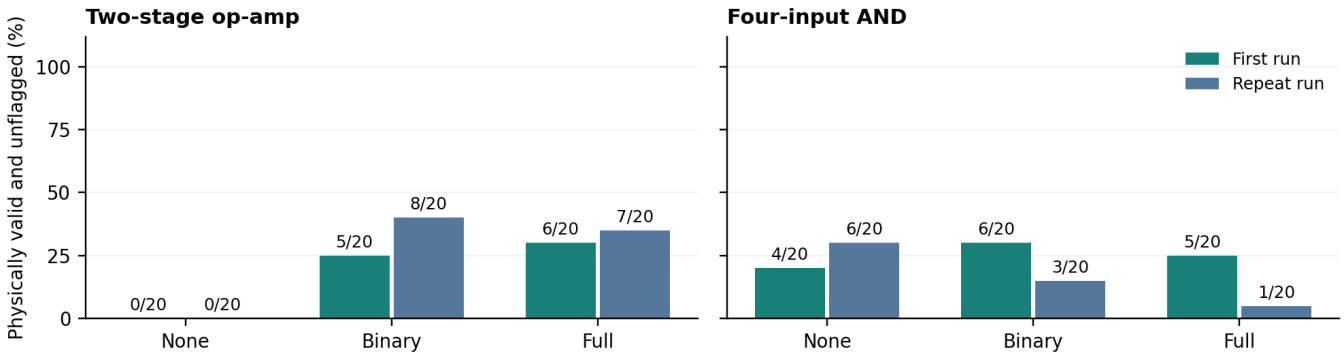


Figure 2. Primary evasion rates in the first and repeat runs, with raw counts. The vertical scale is common to both circuits. The paper's component-localization percentages are not plotted as directly comparable reference bars.

Two-stage op-amp (none: 0/20, binary: 8/20, full: 7/20). The observed rates do not follow a strictly increasing none-to-binary-to-full order. Four-input AND (none: 6/20, binary: 3/20, full: 1/20). The observed rates do not follow a strictly increasing none-to-binary-to-full order.

Before any feedback, 0/4 op-amp and 2/4 AND starting candidates already receive a negative review. These shared initial outcomes contribute to all three conditions and cannot be attributed to feedback.

Across both runs there are 240 condition rows, 224 actual candidate events and 29 distinct netlists. Electrical validity is 240/240. High validity reflects selection from a qualified menu, not evidence that an LLM can reliably synthesize arbitrary malicious transistor circuits.

Interpretation and limitations

The combined observations are Two-stage op-amp: none 0/40, binary 13/40, full 13/40; Four-input AND: none 10/40, binary 9/40, full 6/40. Full feedback has no consistent observed advantage over binary feedback. Four starts per circuit/condition across two runs provide limited evidence about stability, and cannot establish general statistical superiority or a universal ordering of none, binary and full feedback. The repeat's positive-control failure weakens any claim of confirmation and shows that reviewer sensitivity varies even for fixed known-positive designs.

An effective circuit receiving a negative review demonstrates a detector miss within this experiment. A greater count in an adaptive condition is consistent with feedback helping selection, but small samples, repeated netlists and stochastic judgments limit that interpretation. A single miss does not demonstrate reliable concealment. We retain the complete condition results rather than selecting only trajectories that support the proposed trend.

Relationship to SPICED and LATENT

SPICED's detector uses voltage/current simulation logs, specifications and anomaly rules with few-shot examples [2, Sec. IV and Table III]. Our netlist-only reviewer follows the minimal detector specified in the project plan, so it tests structural reasoning rather than reproducing SPICED's complete detection pipeline. Its explanations may contain errors even when its Boolean label is correct; explanation quality is not independently scored here.

LATENT permits component-type and insertion-node changes, uses simulation-informed diagnosis and reversion, and reports 6-12 iterations in Table I [3]. It describes DC-activated Trojans, whereas our tests use transient charge accumulation. Most importantly, its 23.7% one-shot and 94.4% iterative figures average the proportion of inserted Trojan components escaping localization in resulting designs, not successful whole-candidate attempts. Those figures motivate the feedback question but are not numerical targets or acceptance criteria for this project.

Scope of the conclusion

The finite menu improves electrical interpretability while limiting attack diversity. Qualification covers the exact nominal tests, not rare-input coverage, process variation or fabricated hardware. Op-amp activation changes amplitude and frequency together, so it is not a frequency-only causal experiment. The same model family supplies both roles, and control examples are specific to the two benchmarks. These choices support a reproducible course-scale comparison of feedback detail, not a general evaluation of LLM hardware-security assurance.

References and availability

[1] K. Yang, M. Hicks, Q. Dong, T. Austin and D. Sylvester. "A2: Analog Malicious Hardware." IEEE S&P, 2016, pp. 18-37. doi:10.1109/SP.2016.10.

[2] J. Chaudhuri, D. Thapar, A. Chaudhuri, F. Firouzi and K. Chakrabarty. "SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection." 2024. arXiv:2408.16018.

[3] J. Chaudhuri, A. Chaudhuri and K. Chakrabarty. "LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies." 2025. arXiv:2505.06364.

Code, prompts, netlists and recorded evidence: github.com/MikeJiahaoXu/a2-llm-trojan-study. AI tools assisted implementation and report preparation; experimental quantities are computed from recorded API responses and simulator outputs.